FPGA MAC Accelerator: Project Overview

Benchmarking Sequential and Parallel FPGA Multiply-Accumulate Architectures

Objective: measure how the number of parallel MAC units affects latency, throughput, and FPGA resources, without assuming more is better.

Workload

$$Y = \sum_{i=0}^{15} A_i \times B_i$$

- 16-element dot product, 8-bit integers
- Core operation in AI and signal processing

Deliverables

3 verified designs, testbench, $\geq$2 graphs, report, demo

Architectures (same inputs for all)

Design	MACs	Type
Baseline	1	Sequential
Comparison A	2	Partially parallel
Comparison B	4	Parallel

Metrics

- Latency (cycles, µs)
- Throughput (dot products/s)
- Speedup vs. 1 MAC
- DSP, LUT, FF usage
- Max clock (Vivado timing)

Method

- Same inputs and clock
- Outputs checked vs. Python
- $\geq$5 trials per design, raw data kept
- Report mean $\pm$ spread
- Simulated vs. hardware labeled

Platform

- Nexys A7-100T (Artix-7)
- Verilog in AMD Vivado
- Python reference

Four-Week Implementation & Benchmarking Schedule

Technical task	Owner	Week 1 Oct 19–25	Week 2 Oct 26–Nov 1	Week 3 Nov 2–8	Week 4 Nov 9–15
Finalize specs and scope	Both				
Set up Vivado and Nexys A7 target	Ben				
Python reference and test vectors	Christian				
1-MAC baseline and testbench	Christian				
2-MAC and 4-MAC designs	Ben				
Verify outputs against Python	Both				
Synthesis, timing, resource analysis	Ben				
Benchmark trials ($\geq$5 per design)	Both				
Organize data, make graphs	Christian				
Analyze speedup and trade-offs	Both				
FPGA integration, demo prep*	Ben				
Assist FPGA validation	Christian				
Final report, slides, demo	Both				
Milestone		Verified baseline	3 working designs	Benchmark data	Report + demo

Christian Ben Both

*__Backup plan:__ if FPGA integration is difficult, we will use the same Verilog in Vivado simulation and synthesis reports (with instructor approval).

__Potential Extension (If Time Permits):__ test vector lengths 4, 8, 16, 32 to see how workload size affects speedup and efficiency.